# Nowcasting and Temporal Disaggregation of R&D Expenditures

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**Abstract**
Timely macroeconomic data are essential for economic monitoring and policy, but official statistics are often delayed and infrequent. We address this "ragged-edge" problem using a two-step framework applied to gross domestic expenditure on R&D (GERD) series. Step A nowcasts GERD with a neural network using mixed-frequency macroeconomic variables and Internet search volume data. Step B uses model-implied sensitivities and search intensity to allocate annual totals across months. In chronological evaluation, search data alone are competitive with conventional macroeconomic predictors. Once high-dimensional search features enter, the neural network outperforms linear and mixed-frequency benchmarks. Monthly estimates correlate strongly with sparse temporal-disaggregation estimates.

**Keywords:** economic nowcasting; temporal disaggregation; mixed-frequency data; Internet search data; research and development expenditure; machine learning

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# 1 Introduction

Three megatrends are transforming the forecasting literature: the burgeoning of machine learning (ML) methods, the expanding volume of available data, and increasing computing capacity. Together, these trends have shifted empirical research toward prediction-oriented frameworks that emphasize out-of-sample performance and real-time applicability, particularly in policy-relevant settings. They give rise to new classes of models capable of processing larger and more diverse data sources. Notable examples include satellite data (Henderson et al., 2012; Diebold et al., 2021), textual data (Bollen et al., 2011; De Caigny et al., 2020), and price data from online merchants used to measure inflation (Cavallo and Rigobon, 2016). Combined, these developments enable more accurate forecasting and the measurement of economic attributes that were previously difficult to observe (Zhao and Yang, 2023; Ashtiani and Raahemi, 2023; Shu and Ye, 2023). Recent applications illustrate that such prediction-oriented approaches are particularly valuable for macroeconomic measurement tasks in which official statistics are released with delays and subject to revisions, motivating the use of flexible models designed for real-time economic assessment (Zhang et al., 2023; Kaustubh and Ranjan, 2025). More broadly, ML has increasingly been used for prediction tasks in economic policy contexts, ranging from macroeconomic measurement to the forecasting of policy outcomes (Erokhin and Zagler, 2024).

Our paper contributes to this research line. We introduce a framework, illustrated in Figure 1, in response to the so-called "ragged-edge" problem, namely publication delays affecting headline variables in official statistics (Mosley et al., 2022), and the low frequency at which they are collected. The framework consists of a supervised learning task that predicts observed low-frequency figures (*Step A*) and a temporal-disaggregation procedure that estimates unobserved high-frequency figures (*Step B*).

*Step A* is a neural network-based model that incorporates low-frequency data and high-frequency data, in the form of web-search data, to nowcast countries' R&D expenditures. Its primary objective is improving out-of-sample predictive accuracy rather than structural estimation. As innovation is a central driver of economic growth (Romer, 1990; Aghion and Howitt, 2008), R&D investment is an important policy metric. However, it is poorly monitored because official estimates are based on expensive surveys, released yearly or sometimes biennially, and

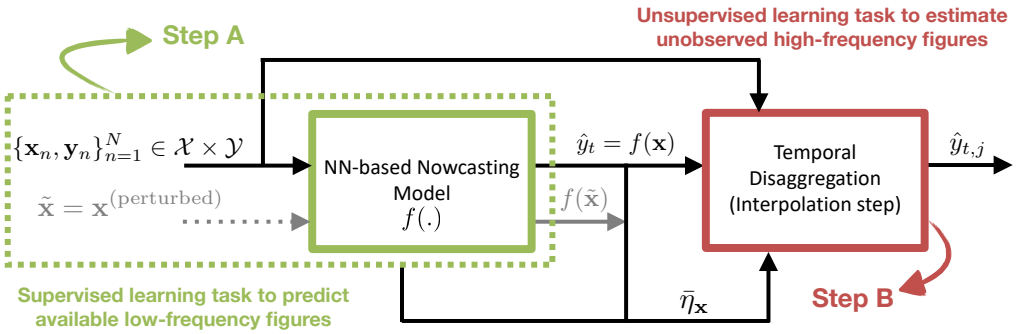



Figure 1: Proposed Framework for Nowcasting R&D Expenditures.

subject to a publication lag of two to three years. This delay hinders effective policy-making. Governments seek to stimulate R&D investment (Edler and Fagerberg, 2017), while institutions monitor countries' innovation performance (WIPO, 2023). The inability to estimate recent R&D investment is therefore particularly consequential (OECD, 2009). It also complicates the monitoring of policy objectives such as the EU target of R&D investment amounting to 3 percent of GDP by 2030 (European Commission, 2020).

*Step B* temporally disaggregates the annual R&D series using the higher-frequency information contained in our input data, allocating observed yearly expenditures into monthly estimates. These estimates can also be aggregated into quarterly series, which provide a more stable basis for monitoring within-year developments and examining patterns around economic shocks (OECD, 2009; Pellens et al., 2024) or policy changes (Guellec and van Pottelsberghe, 2003; González and Pazó, 2008; Tajaddini and Gholipour, 2021). The proposed method is an important step toward nowcasting R&D expenditure at monthly and quarterly frequencies, potentially enabling more timely analysis of innovation dynamics over the business cycle.

The paper makes four contributions. First, we develop a mixed-frequency prediction framework in Step A to nowcast annual GERD that respects ragged-edge information constraints and avoids look-ahead bias. Second, we assess the models chronologically, using earlier country-years for training and later country-years for evaluation. Under this design, Internet search volume data are competitive with traditional macroeconomic predictors for nowcasting annual R&D expenditures. In particular, in the search-only input specification that we also use for temporal disaggregation in Step B, the neural network records a lower MAPE and RMSE and a higher test-set $R^2$ than the linear benchmark using the same feature

set and the mixed-frequency benchmarks constructed from comparable search information. The reported linear benchmark is centered minimum-norm OLS using the complete neural network feature set. The mixed-frequency benchmarks include conventional low-dimensional mixed data sampling (MIDAS) regressions based on a composite search indicator, defined as a single monthly index formed by averaging all search topics. They also include a high-dimensional sparse-group-LASSO MIDAS model that retains all topics observed for every country as separate predictors. Because Step B uses implied sensitivities from the model in Step A, the relevant question goes beyond forecast accuracy. It extends to which prediction models produce useful and stable weights for temporal disaggregation. Third, we introduce a model-implied sensitivity approach that can also accommodate alternative prediction models used in Step A. Fourth, we assess the resulting monthly estimates by comparing them with classical regression-based and sparse temporal-disaggregation methods. We also compare the estimates with independently observed monthly employment in scientific R&D services. Because the true monthly GERD path is unobserved, these comparisons provide evidence about the coherence and economic plausibility of the estimates, but they do not validate the monthly path or rank the estimators. Throughout, the objective is to improve the timeliness of economic measurement and monitoring.

The empirical analysis proceeds as follows. We estimate and evaluate Step A across alternative information sets and forecasting benchmarks. We then apply Step B and assess the resulting monthly estimates using established methods for temporal disaggregation and an external employment series.

The rest of the paper is organized as follows. Section 2 discusses the background literature. Section 3 outlines the neural network-based nowcasting model for yearly frequency. Section 4 describes the data used for the empirical analysis, and Section 5 discusses the results. Section 6 presents the temporal-disaggregation method and its validation. Section 7 concludes and discusses possible extensions.

# 2 Background

Macroeconomic variables play a crucial role in guiding global economic decisions. Covering key facets of economic activity, they offer invaluable insights to policymakers and businesses about an economy's health and direction. Despite their importance, traditional methods of data acquisition have well-known limitations. They rely on expensive surveys, infrequent data collection, and substantial publication delays, which can obscure rapidly changing economic conditions and limit their usefulness for real-time economic assessment.

Acknowledging these limitations, scholars have sought to *nowcast* economic activity. Nowcasting aims to predict the present, the near future, or the recent past, thereby addressing the growing demand for timely and policy-relevant economic information (Banbura et al., 2010). The first generation of nowcasting models consisted primarily of classical statistical and econometric approaches exploiting macroeconomic variables (Evans, 2005; Giannone et al., 2008; Banbura et al., 2010). For instance, the seminal work of Giannone et al. (2008) proposes a dynamic factor model to predict economic activity as captured by GDP. While these models pioneered the nowcasting literature, their reliance on conventional macroeconomic indicators and restrictive parametric forms can limit their ability to use alternative high frequency data and capture more complex predictive patterns.

A subsequent line of inquiry has addressed these limitations by incorporating high-frequency data sources and developing more flexible prediction frameworks. On the data side, a growing literature shows that Internet search activity, such as Google Trends, contains timely information about underlying economic behavior that is not readily captured by conventional indicators and can improve economic forecasting and nowcasting performance (Choi and Varian, 2009, 2012; Wu and Brynjolfsson, 2015; Ferrara and Simoni, 2019; Götz and Knetsch, 2019; Preis et al., 2013; Borup et al., 2023; Woloszko, 2020).[1] This information is particularly valuable when official statistics are affected by publication lags. Beyond Internet search data, high-frequency transaction data can support real-time tracking and nowcasting of private consumption expenditures (Fourné and Lehmann, 2026). In innovation-related settings, Takahashi

[1]Google Trends is an analytical tool that quantifies search intensities for given search terms by geographical area and over selected time periods. See Section 4.2 for details.

et al. (2024) show that search trends can signal innovation diffusion before patent filings and academic publications reach takeoff. Internet-derived data have also been used to measure innovation outputs, including invention commercialization, at scale (Bottai et al., 2025).

In parallel, advances in econometric modeling have led to the development of MIDAS approaches (Ghysels et al., 2004) and unrestricted MIDAS (U-MIDAS) models (Foroni et al., 2015). These frameworks provide principled ways of incorporating predictors observed at different frequencies into macroeconomic analysis. MIDAS models rely on distributed lag polynomials to map high-frequency predictors into low-frequency targets (Ghysels et al., 2004; Clements and Galvão, 2008), while U-MIDAS models relax these restrictions and allow for more direct inclusion of high-frequency information without imposing a specific lag structure (Foroni et al., 2015). Factor MIDAS provides a related approach that combines large information sets with higher-frequency indicators for ragged-edge nowcasting of low-frequency macroeconomic variables (Marcellino and Schumacher, 2010). Recent applications show that such approaches can substantially improve real-time macroeconomic measurement when data arrive asynchronously and with delays (Borup et al., 2023).

More recently, advances in ML have reinforced a broader shift toward prediction-oriented approaches in macroeconomic forecasting and nowcasting. These methods are designed to handle high-dimensional data, nonlinear relationships, and complex interactions, and they are typically evaluated based on out-of-sample predictive performance rather than structural parameter estimation. A growing body of empirical work demonstrates the usefulness of ML models in real-time evaluations and historical evaluations that preserve the order in which information became available, particularly for macroeconomic measurement tasks subject to publication lags and data revisions (Sokolov-Mladenović et al., 2016; Dai et al., 2017; Tümer and Akkuş, 2018; Zhang et al., 2023; Kaustubh and Ranjan, 2025; Borup and Schütte, 2022; Woloszko, 2023; Sun et al., 2025).

Our work also connects to research on R&D dynamics (Bloom, 2007) and the forecasting of R&D and innovation activity. This literature includes traditional econometric techniques (Cheng et al., 2005) and more recent computational methods, including neural networks and text mining, for identifying technological opportunities and mapping innovation dynamics from

large-scale data (Cammarano et al., 2024; Fronzetti Colladon et al., 2025). A related literature develops regression-based methods for temporal disaggregation when target variables are observed only at low frequencies (Chow and Lin, 1971; Mosley et al., 2022).

Despite these advances, limited attention has been given to combining Internet search data with macroeconomic statistics in a single framework that addresses both publication delays and the absence of observed series at higher frequencies. We address this gap through a two-step framework for macroeconomic variables observed with delay and at low frequency, which we apply to annual GERD. The first step nowcasts the observed low frequency statistic using mixed frequency data, including macroeconomic indicators and Internet search volumes from Google Trends. The second step uses model implied predictive sensitivities and search intensity within each year to construct monthly estimates consistent with the annual totals. By considering alternative input configurations, the framework combines mixed frequency information with flexible prediction while prioritizing out of sample accuracy and timely measurement. It also connects annual nowcasting to temporal disaggregation and allows the resulting estimates to be compared with established methods.

# 3 A mixed-frequency neural network-based nowcasting model

## 3.1 General set-up

Our modeling framework is explicitly prediction-oriented and designed for real-time economic measurement rather than structural estimation or policy counterfactual analysis. The objective is to nowcast current levels of R&D expenditures using only information that would have been available at the time of prediction, while respecting publication lags and mixed-frequency constraints. Consequently, model performance is evaluated by fitting the models on earlier observations and testing their predictive accuracy on later observations. The emphasis is on prediction rather than the estimation of parameters intended to represent stable economic relationships. We consider a target ($y \in \mathbb{R}$) sampled at a yearly frequency and a vector

of predictors ($\mathbf{x} \in \mathbb{R}^d$) containing mixed-frequency inputs, *i.e.*, some variables are sampled monthly and others yearly. The pair $(\mathbf{x}, y)$ represents an observation for which the general prediction model is given by the reduced-form relationship:

$$y = f(\mathbf{x}) + \varepsilon$$

in which $\varepsilon$ is a zero-mean error term. The target value denotes the R&D expenditures for each country $i$ at year $t$, and the input vector ($\mathbf{x}$) consists of four main components. We categorize these components into the following vectors: autoregressive (AR) terms of the target ($\mathbf{Y}_{t-\tau,i} \in \mathbb{R}^{\tau}$), search volume data, *i.e.* Google trends data ($\mathbf{S}_{t,j,i} \in \mathbb{R}^{(1+\tau)\cdot k_s}$), macroeconomic variables ($\mathbf{Z}_{t-\tau,i} \in \mathbb{R}^{\tau \cdot k_z}$), and the general categorical features ($\mathbf{C}_{j,i} \in \mathbb{R}^{k_c}$), as defined further below. We can re-write the prediction model as:

$$y_{t,i} = f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{S}^{(m)}_{t,j,i},\ \mathbf{Z}_{t-\tau,i},\ \mathbf{C}^{(m)}_{j,i};\ \theta^{(m)}\right) +\ \varepsilon^{(m)}_{t,j,i}. \tag{1}$$

This model accounts for variations that occur on a monthly basis thanks to the inclusion of monthly-level features in the predictor set, as indicated by superscript $(m)$. Note also the presence of the superscript on the model $f(.)$ itself, highlighting the fact that the model contains features with a different sampling frequency than the target value. Finally, $\theta^{(m)}$ represents the corresponding vector of Step A model parameters. The mixed-frequency structure is designed to approximate the information set available to analysts and policymakers. The Google trends variables use either lagged annual averages or current-year observations available through the preceding month, so they do not introduce forward-looking information. The chronological split also prevents future country-years from entering model estimation. A separate publication-vintage qualification applies to configurations containing lagged GERD: because official GERD is released with a delay of two to three years, the first and sometimes second annual lags may not yet be observable in a strict real-time application. These richer configurations are therefore informative comparisons of predictive content rather than fully publication-lag-feasible specifications; the search-only *AGT* and *MGT* configurations do not use lagged GERD.

We now discuss each of the four components of the model. The vector $\mathbf{Y}_{t-\tau,i}$ includes $\tau > 0$ lagged values of the target to account for potential serial correlation in $y_{t,i}$, without adding any forward-looking (look-ahead) bias:[2]

$$\mathbf{Y}_{t-\tau,i} = \begin{bmatrix} y_{t-1,i} & \cdots & y_{t-\tau,i} \end{bmatrix}^{\top}.$$

In the model-ready panel, these lag variables refer to preceding available GERD observations for a country. Accordingly, for countries with irregular reporting, $t-\ell$ in the lag notation denotes the $\ell$-th preceding available observation and does not necessarily denote calendar year $t-\ell$. An observation lag can therefore span more than one calendar year. Remaining undefined lag entries are set to zero before the training-fitted feature standardization is applied. Missingness indicators are retained in the prepared dataset for auditing. Configurations containing GERD lags should consequently be interpreted as information-rich predictive comparisons rather than strictly publication-vintage-feasible models.

The second vector of predictors, $\mathbf{S}^{(m)}_{t,j,i}$, consists of monthly data obtained from Google trends. The data contain $k_s$ number of different topics associated with a selection of search terms, as we elaborate in Section 4. Note the subscript $j$, which indicates values corresponding to the $(12-j)^{th}$ month of the year, with $j = 0, ..., 11$.[3] Formally, we define $\mathbf{S}^{(m)}_{t,j,i}$ as follows:

$$\mathbf{S}^{(m)}_{t,j,i} = \begin{bmatrix} \bar{\mathbf{S}}_{t-\tau,i} & \bar{\mathbf{S}}^{\mathrm{YTD}^{(m)}}_{t,j,i} \end{bmatrix}^{\top}.$$

The first sub-vector, $\bar{\mathbf{S}}_{t-\tau,i}$, contains the lagged monthly Google trends values averaged on a yearly frequency for the past $\tau$ years and for $k_s$ number of predictors representing various topics associated with the selected search terms. We define:

$$\bar{\mathbf{S}}_{t-\tau,i} = \begin{bmatrix} \bar{\mathbf{s}}_{t-1,i} & \cdots & \bar{\mathbf{s}}_{t-\tau,i} \end{bmatrix},$$

[2] We set $\tau = 3$ throughout the empirical analysis. Section 5 summarizes, and Appendix I.E.1 reports in detail, a lag-order sensitivity check for the search-only specification used in temporal disaggregation.

[3] Considering the scalars $s_{k_s,t,j,i}$ with $j$ corresponding to the $(12-j)^{th}$ month of the year $t$ for country $i$, is equivalent to $s_{k_s,t-j/12,i}$. It would be interpreted as follows: December $s_{k_s,t-0/12,i}$, November $s_{k_s,t-1/12,i}$, ... , June $s_{k_s,t-0.5,i}$, ... , and January $s_{k_s,t-11/12,i}$.

$$\bar{\mathbf{s}}_{t-\tau,i} = \begin{bmatrix} \bar{s}_{1,t-\tau,i} & \dots & \bar{s}_{k_s,t-\tau,i} \end{bmatrix},$$

$$\bar{s}_{k_s,t-\tau,i} = \frac{1}{12} \sum_{j=0}^{11} s^{(m)}_{k_s,t-\tau,j,i}.$$

By aggregating the Google trends data into annual average for each topic $\bar{s}_{k_s,t-\tau,i}$ and constituting $\bar{\mathbf{s}}_{t-\tau,i}$ for all topics, we reduce the monthly variability and emphasize long-term trends. Also, by constructing the time-lagged vector $\bar{\mathbf{S}}_{t-\tau,i}$ for $\tau$ lagged values of these averages, we enable the model to capture temporal dynamics across several years and test their associations with the future outcomes.

The second sub-vector of $\mathbf{S}^{(m)}_{t,j,i}$, which is $\bar{\mathbf{S}}^{\text{YTD}^{(m)}}_{t,j,i}$, captures the year-to-date (YTD) Google trends values averaged over the past months of the current year (excluding the current month):

$$\bar{\mathbf{S}}^{\text{YTD}^{(m)}}_{t,j,i} = \begin{bmatrix} \bar{s}^{\text{YTD}^{(m)}}_{1,t,j,i} & \dots & \bar{s}^{\text{YTD}^{(m)}}_{k_s,t,j,i} \end{bmatrix},$$

$$\bar{s}^{\text{YTD}^{(m)}}_{k_s,t,j,i} = \begin{cases} \frac{1}{11-j} \sum_{j'=j+1}^{11} s^{(m)}_{k_s,t,j',i} & \forall j \in [0, 10], \\ 0 & \forall j = 11. \end{cases}$$

In short, while the subvector $\bar{\mathbf{S}}_{t-\tau,i}$ captures the information from previous year(s), the subvector $\bar{\mathbf{S}}^{\text{YTD}^{(m)}}_{t,j,i}$ captures the information available in the current year ($t$) up to the previous month that is being estimated. Indeed, this definition implies that for January ($j = 11$), we have $\bar{s}^{\text{YTD}^{(m)}}_{k_s,t,j,i} = 0$.

The third vector of predictors, $\mathbf{Z}_{t-\tau,i}$, contains the lagged values of $k_z$ different macroeconomic variables. We formally define it as:

$$\mathbf{Z}_{t-\tau,i} = \begin{bmatrix} \mathbf{z}_{t-1,i} & \dots & \mathbf{z}_{t-\tau,i} \end{bmatrix}^\top,$$

$$\mathbf{z}_{t-\tau,i} = \begin{bmatrix} z_{1,t-\tau,i} & \dots & z_{k_z,t-\tau,i} \end{bmatrix}.$$

Finally, the fourth component, $\mathbf{C}_{j,i}^{(m)}$, represents the country and month information, which we discuss further in Section 3.2:

$$\mathbf{C}_{j,i}^{(m)} = \begin{bmatrix} e_{1,i} & \dots & e_{4,i} & m_1^{(m)} & \dots & m_{12}^{(m)} \end{bmatrix}^\top.$$

The month indicators $m_1^{(m)}, \dots, m_{12}^{(m)}$ are explicit one-hot inputs. In contrast, $e_{1,i}, \dots, e_{4,i}$ are not observed country-level predictors supplied to the model. The country label is first mapped to an integer index, and the network uses this index to retrieve a four-dimensional embedding vector from a trainable embedding matrix. The embedding values are therefore learned jointly with the other neural network parameters and enter the nonlinear branch of the model.

As a baseline, expressing the model in equation (1) considering a classical regression setup, in which the input and output data satisfy a linear relation, yields:

$$y_{t,i} = \mathbf{S}_{t,j,i}^{(m)^\top} \omega_S^{(m)} + \mathbf{Y}_{t-\tau,i}^\top \omega_{AR}^{(m)} + \mathbf{Z}_{t-\tau,i}^\top \omega_Z^{(m)} + \mathbf{C}_{j,i}^{(m)^\top} \omega_C^{(m)} + \varepsilon_{t,j,i}^{(m)},$$

and the vector of model parameters is defined as:

$$\theta^{(m)} = \begin{bmatrix} \omega_S^{(m)^\top} & \omega_{AR}^{(m)^\top} & \omega_Z^{(m)^\top} & \omega_C^{(m)^\top} \end{bmatrix}^\top.$$

We depart from this linear regression model by developing a neural network architecture from the multilayer perceptron (MLP) class (Goodfellow et al., 2016). In this approach, the function $f(.)$ is represented by a neural network that will learn weights and biases through the course of training for a suitable representation of the data. This problem falls under the class of supervised learning. We observe some data (train set) $S_{\text{train}} = \{\mathbf{x}_n, y_n\}_{n=1}^N \in \mathcal{X} \times \mathcal{Y}$ and given a new $\mathbf{x}$, we aim to predict its label $y$, corresponding to R&D expenditures.

Our primary specification is a neural network because it can flexibly use a large and varied set of inputs without imposing a restrictive functional form. This choice is consistent with Radhakrishnan et al. (2022), who emphasize the value of feature learning for capturing complex empirical patterns. Our empirical claim is more limited. The neural network allows for nonlinear relationships and interactions among topics, countries, months, and macroeconomic controls, but the small annual sample does not establish that such relationships are necessary.

We therefore use the neural network as the primary model in Step A and compare it with unregularized and regularized linear alternatives. A suitable Step A model must perform adequately in predicting annual GERD out of sample and produce stable model implied sensitivities for temporal disaggregation.

We adopt a series of configurations for modeling $f(\cdot)$ by varying the input space, and noting that all configurations contain features associated with country and month, $\mathbf{C}_{j,i}^{(m)}$. We start by considering two 'conventional' input spaces. The configuration with the minimal dimensional input space includes only autoregressive terms, referred to as *LagRD* model. Then, extending the set of predictors to macroeconomic variables and autoregressive terms, leads to a configuration labeled as the *Macros* model. Next, we consider an input space that contains exclusively Google trends data and we explore two configurations. Model *AGT* includes the annual Google trends data from previous years, so that $\mathbf{S}_{t,i} = [\bar{\mathbf{S}}_{t-\tau,i}]$. Model *MGT* integrates the yearly data with the monthly Google trends data from the most recent months, that is $\mathbf{S}_{t,i} = [\bar{\mathbf{S}}_{t-\tau,i} \quad \bar{\mathbf{S}}_{t,i}^{\text{YTD}}]$. We then expand the input space by including autoregressive terms of R&D expenditures as additional predictors, hereafter referred to as the *AGTwRD* and *MGTwRD* models, respectively. Finally, we consider the broadest possible input space by combining macroeconomic variables, historical target values, and Google trends data at both yearly and monthly intervals, hereafter called *AllVar*. This model is the general configuration introduced in equation (1). By varying the input space, we will be able to assess the improvements in predictive accuracy that the various pieces of data offer. Table 1 summarizes the seven configurations that we will explore.

The next section elaborates on the MLP set-up to model the function $f(.)$.

## 3.2 MLP set-up

This subsection describes the neural network architecture used to operationalize the prediction function $f(\cdot)$ introduced in Section 3.1. The architectural choices are guided by the objective of real-time nowcasting under mixed-frequency data constraints, rather than by structural interpretability or causal identification. Accordingly, the focus is on flexible function approximation, stable out-of-sample performance, and robustness to high-dimensional input spaces,

Table 1: Summary of Model Configurations

| Configuration | Prediction function given an input space |
|---|---|
| *LagRD* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *Macros* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{Z}_{t-\tau,i},\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *AGT* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{S}_{t,j,i}(\bar{\mathbf{S}}_{t-\tau,i}),\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *MGT* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{S}_{t,j,i}^{(m)}(\bar{\mathbf{S}}_{t-\tau,i},\ \bar{\mathbf{S}}_{t,j,i}^{\mathrm{YTD}^{(m)}}),\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *AGTwRD* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{S}_{t,j,i}(\bar{\mathbf{S}}_{t-\tau,i}),\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *MGTwRD* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{S}_{t,j,i}^{(m)}(\bar{\mathbf{S}}_{t-\tau,i},\ \bar{\mathbf{S}}_{t,j,i}^{\mathrm{YTD}^{(m)}}),\ \mathbf{C}_{j,i}^{(m)}\right)$ |
| *AllVar* | $f(\mathbf{x}) := f^{(m)}\left(\mathbf{Y}_{t-\tau,i},\ \mathbf{S}_{t,j,i}^{(m)}(\bar{\mathbf{S}}_{t-\tau,i},\ \bar{\mathbf{S}}_{t,j,i}^{\mathrm{YTD}^{(m)}}),\ \mathbf{Z}_{t-\tau,i},\ \mathbf{C}_{j,i}^{(m)}\right)$ |

which are central requirements in prediction-oriented economic measurement tasks.

As mentioned in Section 3.1, unlike non-feature learning algorithms that require explicit programming for feature extraction, neural networks autonomously learn and utilize features inherent in the data (Radhakrishnan et al., 2022). Neural networks provide flexible, high-capacity function approximators that are well suited to settings with nonlinearities, interactions, and heterogeneous predictors, while still exhibiting strong generalization properties when evaluated out of sample (Belkin, 2021). In this subsection, we present the neural networks set-up from the MLP class developed for our context.

We model the prediction function $f(\cdot)$ using a feed-forward neural network from the multilayer perceptron (MLP) class. The network maps the input vector $\mathbf{x} \in \mathbb{R}^d$ to a scalar prediction through a sequence of fully connected hidden layers with nonlinear activation functions, followed by a linear output layer. As this architecture is standard in the machine learning literature, we refer to Goodfellow et al. (2016) for formal definitions and focus here on the elements relevant for our empirical application.

For completeness, we define $f : \mathbb{R}^d \to \mathbb{R}$ as a fully connected network with $L$ hidden layers for $L > 1$, weight matrices $\{\mathbf{W}^{(l)}\}_{l=1}^{L+1}$, bias vectors $\{b^{(l)}\}_{l=1}^{L+1}$, and activation function $\phi$, of the form:

$$f(\mathbf{x}) = \mathfrak{h}(\mathbf{x})^\top \mathbf{W}^{(L+1)} + b^{(L+1)};$$

$$\mathbf{x}^{(\ell)} = h^{(\ell)}\left(\mathbf{x}^{(\ell-1)}\right) := \phi\left(\mathbf{W}^{(\ell)^\top}\mathbf{x}^{(\ell-1)} + b^{(\ell)}\right) \quad \text{for } \ell \in \{2, \dots, L\};$$

$$\mathfrak{h}(\mathbf{x}) := \mathbf{x}^{(L)} = h^{(L)}\left(\mathbf{x}^{(L-1)}\right) = \phi\left(\mathbf{W}^{(L)^\top}\mathbf{x}^{(L-1)} + b^{(L)}\right)$$

with $\mathbf{x}^{(0)} = \mathbf{x}$ for $\ell = 1$, and $\mathfrak{h} : \mathbb{R}^d \to \mathbb{R}^K$. The function $\mathfrak{h}(.)$ represents the feature extractor part of the architecture, with activation function $\phi$ that performs an element-wise non-linear transformation on its input. In economic terms, this feature extraction stage allows the model to construct latent representations of the mixed-frequency inputs that summarize complex statistical dependencies across countries, over time, and across heterogeneous information sources, before mapping them to the target outcome.

The model learns its parameters by minimizing a Huber loss on the transformed target. For estimation, the target is transformed to a within-country log-standardized scale. For each country $i$, we center and scale its log R&D expenditure using the mean $\mu_i$ and standard deviation $\sigma_i$ computed over that country's *training* years, $z_{t,i} = (\log y_{t,i} - \mu_i)/\sigma_i$, and map predictions back to levels for reporting as $\hat{y}_{t,i} = \exp(\hat{z}_{t,i}\,\sigma_i + \mu_i)$ (here and throughout, log denotes the natural logarithm).[4] This transformation stabilizes training and removes the country-specific level and scale, allowing the network to focus on predictive variation; because it introduces only country-specific constants estimated on the training sample, it does not add a lagged target value as a feature (Section 5). All continuous predictors are likewise standardized using means and standard deviations computed on the training sample only.

The implemented model has two fully connected hidden layers with 64 and 16 neurons. Each hidden layer applies layer normalization, a sigmoid linear unit (SiLU) activation, and dropout with rate 0.12. A four-dimensional learned country embedding is concatenated with the standardized numerical predictors and the 12 one-hot month indicators before entering the nonlinear branch. In addition, a direct linear skip connection maps the standardized numerical predictors and month indicators to the output. The final prediction on the transformed scale is the sum of this linear component and the output of the nonlinear branch. This wide-and-deep structure nests a regularized linear contribution while allowing nonlinearities and interactions when supported by the data.

[4]The constants $\mu_i$ and $\sigma_i$ are estimated once on each country's training years and then reused for its validation and test observations; no test-period information enters their computation, so the transformation introduces no look-ahead. It does, however, require that each target country have a training history so that $\mu_i$ and $\sigma_i$ are available, which is the standard nowcasting setting in which recent years are predicted for a country with historical data.

We screened alternative pure neural network architectures and training hyperparameters for the *AGT* specification under the temporal evaluation design, without residual learning or additional feature engineering. The selected specification was then held fixed across all seven input configurations so that their reported comparison reflects changes in the information set rather than configuration-specific architecture tuning.

In order to diminish sensitivity to random initialization, we construct an ensemble of fifteen independently initialized neural networks and average their predictions. This approach stabilizes the model's performance by combining predictions from multiple neural networks (Woloszko, 2020).

Furthermore, as mentioned in Section 3.1, the vector of predictors $\mathbf{C}_{j,i}^{(m)}$ contains features associated with months and countries. As for months, we implement 'one-hot encoding,' treating each calendar month as a separate categorical feature. These month indicators act as a fixed categorical seasonality control in the annual prediction model: they allow the network to learn average calendar-month effects in the mixed-frequency input rows. Because the model is nonlinear, these effects are learned through the network weights and interactions with other predictors rather than as a single interpretable month-specific regression coefficient; imposing additional month-specific coefficients or separate month-by-country effects would consume degrees of freedom quickly and risks overfitting in our short annual panel. Country identifiers are first mapped to integer indices, which are then passed to an 'embedding layer' that learns a four-dimensional continuous representation for each country. This approach allows the model to capture latent similarities across countries in a data-driven manner and is commonly used in neural network models with categorical inputs (Guo and Berkhahn, 2016).[5] As training progresses, countries exhibiting similar predictive patterns may obtain embedding vectors that are closer in the latent space, reflecting shared characteristics relevant for the nowcasting task.

In terms of model scope, Woloszko (2020) discuss the trade-offs between country-specific and cross-country modeling. While country-specific models can capture fine-grained heterogeneity, they require richer data availability. Cross-country pooling, by contrast, increases the

[5]The indexing step is a deterministic preprocessing operation used solely to interface with the embedding layer and does not imply ordinal or numerical meaning. The learned embeddings, rather than the indices themselves, enter the prediction function.

effective sample size and improves estimation robustness, reflecting a bias-variance trade-off. Given the prediction-oriented objective and the relatively short time dimension of the data, we adopt a pooled cross-country specification.

Layer normalization is applied to the output of each hidden linear transformation before the SiLU activation. Unlike batch normalization, this operation normalizes within each observation and does not depend on batch-level moments, which is useful in the present small-sample setting.

Model parameters are optimized using AdamW with a learning rate of $3 \times 10^{-3}$ and weight decay of $5 \times 10^{-4}$. AdamW extends the Adam algorithm by decoupling weight decay from gradient-based updates (Loshchilov and Hutter, 2017). Training uses mini-batches of 64 observations and clips the gradient norm at 5 to limit unstable updates.

Each ensemble member is first trained for at most 500 epochs with early stopping based on annual validation MAPE after predictions are mapped back to GERD levels. The checkpoint with the lowest validation MAPE is retained, and training stops after 65 consecutive epochs without improvement. This stage selects the hyperparameters and training duration. With these choices fixed, each model is reinitialized and trained on the combined training and validation periods to estimate its weights. All preprocessing parameters are recomputed on this combined sample, and the model is evaluated on the untouched test period. The centered minimum-norm OLS and Ridge comparisons use the same estimation sample and preprocessing information. Taken together with the chronological validation/test split, fit-sample standardization, Huber loss, weight decay, layer normalization, dropout, gradient clipping, and ensembling, this design mitigates the main overfitting risks created by the high-dimensional predictor set. Nevertheless, given the small number of annual country-year observations, these safeguards cannot eliminate small-sample uncertainty; we therefore interpret close model rankings and forecast-comparison tests cautiously. To preserve anonymity during peer review, the code and data repository is withheld and will be made publicly available upon acceptance.

# 4 Data

This section provides an overview of the data underpinning the empirical application. The target variable is GERD, which is published annually by the OECD. As explained in Section 3, we generate GERD nowcasts using mixed-frequency data, consisting of macro variables and Google trends data. In this study, we focus on eight selected countries out of the top twenty innovative countries, according to WIPO's Global Innovation Index (GII) (WIPO, 2023). The selected countries in our sample cover different continents, including Switzerland, the United States, the United Kingdom, Germany, South Korea, China, Japan, and Canada.

## 4.1 GERD

GERD measures the total expenditures (both current and capital) on R&D activities conducted by all organizations within a nation's territory, including companies, research institutes, universities, and government laboratories (OECD, 2023). It includes R&D funded from external sources but excludes domestic funds designated for R&D activities outside the domestic economy. These data are primarily gathered through surveys conducted by national authorities and then reported to the OECD (OECD, 2015). National authorities follow the international norms set forth by the Frascati Manual (OECD, 2015). The R&D series are expressed in billions of 2015 USD PPPs to ensure comparability over time and across countries.

The GERD series have data gaps and irregular reporting intervals for certain countries, particularly Switzerland. We retain only observed GERD values as prediction targets. In the model-ready panel, lagged GERD predictors are formed from preceding available observations within each country, so an observation lag need not equal one calendar year when reporting is irregular. Undefined lag entries are set to zero before feature standardization. A binary indicator records whether the original GERD figure is missing for data-auditing purposes.

## 4.2 Google trends

Google trends, developed by Google, measures search intensities for any given queried keyword(s) by geographical area and over any selected time period starting in 2004 (Google News

Initiative, 2023). The tool provides normalized data on the relative search volume of queries. Each data point in Google trends is normalized by the total search volume of the geography and time range considered, on a scale between 0 and 100. The search volume index (SVI) for a given search term or topic $k_s$ is denoted as $s^{(m)}_{k_s,t,j,i}$ and defined as follows:

$$s^{(m)}_{k_s,t,j,i} = \frac{\mathrm{SV}^{(m)}_{k_s,t,j,i}}{\mathrm{TSV}^{(m)}_{t,j,i}} * c_{i,k_s} \in [0, 100]$$

in which $\mathrm{SV}^{(m)}_{k_s,t,j,i}$ and $\mathrm{TSV}^{(m)}_{t,j,i}$ stand for volume of searches for $k_s$ in country $i$ at a given time (year $t$ and month $j$), and total number of searches in country $i$ at the selected time range, *i.e.* from 2004 to the given time (year $t$ and month $j$), respectively. The SVI is normalized and also indexed on a scale of 0 to 100 by the constant $c_{i,k_s}$ for a given time series $s^{(m)}_{k_s,t,j,i}$.

We use Google trends data to predict R&D expenditures dynamics with a fine level of granularity. To keep the relevant search terms that might correlate with R&D activities, we start by reconstructing the ecosystem of stakeholders associated with R&D activities. This ecosystem comprises entities that either directly perform R&D or play a supporting role in the broader R&D landscape. We identify the various stakeholders and the specific search terms by studying the 'systems of innovation' literature (Edquist, 2013; Lundvall, 2010). Table 2 presents the ecosystem developed for this study, containing the finalized selection of stakeholders and the corresponding search terms that we identified.[6]

Table 2: Stakeholders and their respective search terms for R&D expenditure.

| Stakeholder | Search Terms |
|---|---|
| Businesses | R&D Expenditure, Product Development |
| Consulting Firms | Innovation Management, Innovation Strategy |
| Government Agencies | Government Grants, Research Funding |
| Innovation Hubs | Startup Incubation, Technology Park |
| Patent Attorneys | Patent Attorney, Patent Registration |
| R&D Employees | R&D Jobs |
| Research Institutions | Collaboration with Industry, Research Grant |
| Tax Authorities | R&D Tax Credit |
| Venture Capitalists (VCs) | VC Investment, Startup Funding |

[6]The initial ecosystem developed for this study encompassed a broader range of search terms, detailed in the appendix, Table I.A.1.

After identifying specific search terms, we extract the related topics from Google Trends. Google Trends groups searches into standardized topics associated with individual search terms. These topics can include the exact terms, common misspellings, and acronyms, and they are harmonized across languages (Google News Initiative, 2023). For example, the search term 'R&D expenditure' is associated with topics including *'Research and development', 'Innovation', 'Technology', and 'Patents'*. Using topics reduces the ambiguity of relying on individual search terms and makes the analysis more comparable across countries with different languages (Woloszko, 2020). Because Google Trends does not provide a fixed list of topics, selecting the relevant topics requires careful exploration. We use the specific search terms in Table 2 as anchors for extracting the associated topic identifiers from Google Trends. In total, we identify 57 Google Trends topics, so $k_s = 57$ in our application.

One feature of Google trends is that search indices are derived from a sample of total search volume, for computational tractability reasons. Indices that represent low-volume searches may exhibit significant sampling variance (Woloszko, 2020). Accordingly, in order to minimize sampling variance associated with Google's methods, we retrieve five samples per query and average search volume index across the samples for each topic to obtain a more reliable time series (Woloszko, 2020; Medeiros and Pires, 2021; Woloszko, 2023).

## 4.3 Macroeconomic variables

The set of predictors also contains six 'traditional' macroeconomic variables capturing elements of countries' wealth, economic conditions, and competitiveness ($k_z = 6$). These variables, which correlate with R&D expenditures, include:

- Gross domestic product per capita: Reflects the wealth and economic strength of an economy.
- Unemployment rate: Indicates the health of the labor market and broader economic conditions.
- Population: Used as a scale variable, represents the size of an economy.
- Inflation rate: Represents the rate of price growth and indicates the state of the economy.

- Export and import volumes: Chosen as indicators of a nation's competitiveness and its engagement with the global economy.

We sourced these variables from the International Monetary Fund (IMF)'s World Economic Outlook Database, April 2023 release (International Monetary Fund, 2023). When macroeconomic variables are used as model inputs, missing values are filled with the mean of the respective variable for that country, calculated over all available years in the complete prepared panel. The chronological split prevents later country-years from entering model fitting. However, the imputation uses the complete prepared panel rather than only information available at each historical forecast date. Configurations containing macroeconomic variables should therefore be interpreted as comparisons of predictive content rather than strict real-time forecasts. A fully operational exercise would estimate the imputation parameters using only information available at each forecast date.

# 5 Results

## 5.1 Out-of-sample predictions

As explained earlier, we present the performance of the MLP setup over different configurations of the input space. By varying the input space, we aim to understand the value of the Google trends data compared to traditional economic variables. The models are assessed chronologically using country-year observations. Within each country, the earliest 64% of years are used for training, the next 16% for validation, and the most recent 20% for testing. After model selection, the training and validation periods are combined to estimate the reported models, while the test period remains unchanged. This split prevents later country-years from entering model estimation and follows the principle that forecasting evaluations with panel data should preserve temporal order (Cerqua et al., 2026). Although the evaluation preserves chronological order, it does not fully recreate real-time data availability. Some preprocessing relies on a fixed version of the data, and some GERD-lag configurations contain information that was unavailable at certain historical forecast dates. Because the test years lie strictly in the future of the training sample and GERD follows a strong upward trend, we predict the target

on a *within-country log-standardized* scale, each country's log GERD is centered and scaled using its training-period mean and standard deviation, and predictions are mapped back to levels for evaluation.[7] Each country-year is represented by twelve monthly input observations, all associated with the same observed annual GERD value. The model produces an annual GERD prediction for each observation, and we average these twelve predictions to obtain one prediction per test country-year before computing the evaluation metrics. This procedure summarizes performance across the within-year rows; it does not represent a forecast made at one particular month of the year.

Figure 2 reports the distribution of annual out-of-sample absolute percentage errors and the root mean squared error (RMSE) for the MLP and centered minimum-norm OLS across the seven input spaces. For the OLS, the predictors are centered internally, and the regression is estimated without a penalty. Because the high-dimensional design matrix is rank deficient, the OLS coefficient vector is not unique. The figure reports the centered minimum-norm solution under the stated parameterization.

First, for every configuration that incorporates the high-dimensional Google Trends feature space (*AGT*, *MGT*, *AGTwRD*, *MGTwRD*, and *AllVar*), the neural network outperforms the reported centered minimum-norm OLS by a wide margin: test MAPE falls from roughly 17.6-23.2% under this OLS implementation to 6.8-9.7% under the neural network, and the out-of-sample $R^2$ rises to 0.96-0.98. This unregularized linear implementation overfits the large, correlated search-topic feature set and extrapolates poorly to the future test years, whereas the neural network provides a more stable prediction function in this high-dimensional setting. This comparison should not be read as evidence that nonlinearity is empirically required; rather, it shows that regularization or flexible feature learning is needed once many correlated search topics enter the model. In the low-dimensional configurations, the two specifications perform similarly for *Macros*, while the neural network performs substantially better for *LagRD*.

[7] This transformation is standard for trending series and addresses the non-stationarity that would otherwise penalize models lacking an autoregressive anchor; importantly, it introduces only a country-specific location and scale constant estimated on the training years, so no lagged value of the target enters as a feature and the "Google-trends-only" interpretation of the *AGT* and *MGT* configurations is preserved. As robustness checks, Appendix I.K reports results under a country-year grouped split (a pooled, non-temporal design) and documents the information leakage that arises under a naïve observation-level randomized split; the qualitative conclusions are broadly unchanged.

The *LagRD* configuration contains no high-frequency search information from the year being predicted and serves mainly as a persistence benchmark. The neural network is also markedly more *stable* than the reported centered minimum-norm OLS: this pattern is also evident under the country-year grouped split (Appendix I.K, Figure I.K.1), where the OLS implementation exhibits much greater error dispersion for every Google Trends configuration, whereas the neural network's errors remain relatively concentrated.

Second, among the neural network models, the search-only *AGT* configuration records a lower test MAPE and a higher test-set $R^2$ than the model based on traditional macroeconomic predictors (*AGT*: 7.25% MAPE, $R^2_{\text{TSS}} = 0.97$; *Macros*: 8.54%, $R^2_{\text{TSS}} = 0.96$). The *MGT* configuration, which also uses search data alone, remains competitive at 9.65% MAPE and $R^2_{\text{TSS}} = 0.96$. The results show that when the models are fitted on earlier country-years and evaluated on later ones, search-volume data can be as informative as conventional macroeconomic indicators, depending on the specification and metric. Adding lagged GERD to *AGT* improves MAPE modestly to 6.82%, while combining all predictors does not improve further (*AllVar*: 8.15%).

As a sensitivity check conducted during model selection, we re-estimate the search-only *AGT* neural network using cumulative lag orders from one through four. Test MAPE decreases with each additional lag, from 18.33% with one lag to 15.34%, 13.96%, and 13.10% with two, three, and four lags, respectively. We retain three lags because they capture most of the observed improvement without expanding the search input from 171 to 228 features. This choice balances predictive accuracy against data requirements, since longer histories may not be available in shorter samples. The lag order can therefore be adapted to the available data, with a possible loss of accuracy when fewer lags are used. Appendix I.E.1 reports the full results. These values come from the model-selection stage and therefore differ from the results reported above.

Overall, the observations of Figure 2 highlight the potential of high-frequency search data for timely R&D measurement. In particular, neural network configurations exploiting search-volume data are competitive with configurations relying solely on conventional predictors, while the neural network consistently outperforms the reported centered minimum-norm OLS whenever the high-dimensional search feature space is included. As a complementary

robustness exercise, we apply the same architecture and training recipe under a country-year grouped split (Appendix I.K). This non-temporal design evaluates interpolation across unseen country-years rather than chronological extrapolation. Under this split, *AGT* attains a MAPE of 5.4%, compared with 5.3% for the macro-based neural network, 50.9% for centered minimum-norm OLS using the same *AGT* inputs, and 19.7% for MIDAS after grouped training-only normalization. The stronger performance under the grouped split provides additional evidence that the search data contain predictive information. The temporal split remains the primary evaluation design because it tests the models on years later than those used for training.

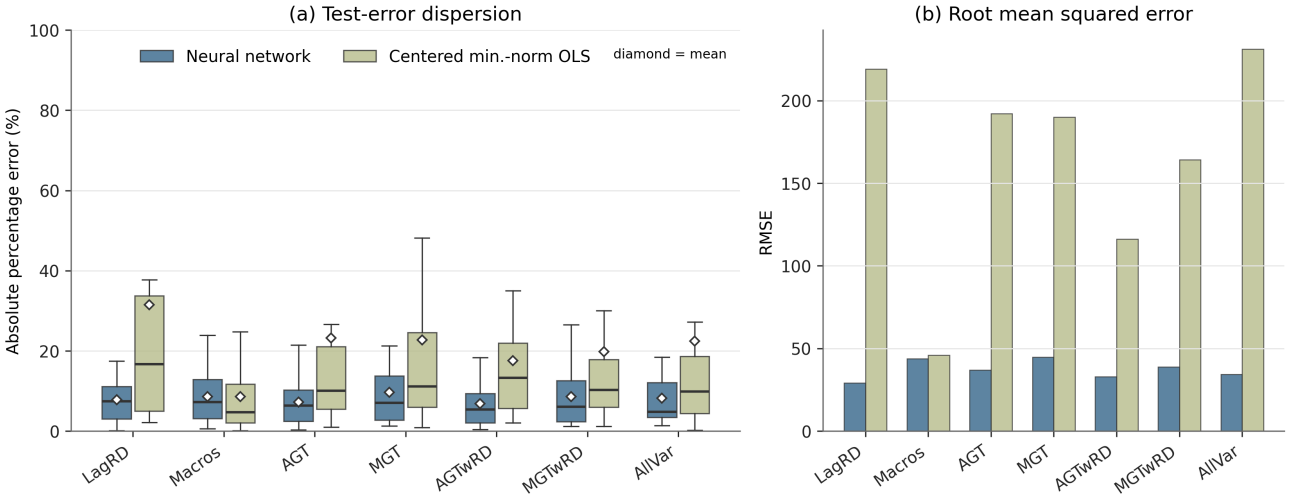



Figure 2: Temporal-split errors by configuration. Panel (a) shows annual absolute percentage error; panel (b) reports RMSE in billions of 2015 USD PPP. The reported models are estimated on the combined training and validation periods and evaluated on the unchanged test period.

The out-of-sample performance of the *AllVar* neural network during model selection is shown in a pooled plot of predicted and observed values in Figure I.E.1 of Appendix I.E. The configuration-level panels in Appendix I.D show the corresponding trajectories for each country. These figures show that performance varies across countries and includes some outlier predictions. These errors may reflect country-specific factors and limitations in the representativeness of Google Trends data, particularly in its early years or in countries where Google searches are less widely used.

Regarding Switzerland, the R&D expenditures data are reported biennially since 2015 and prior to that, only once every four years. This irregular frequency and sparse data availability can hinder the model's capability to accurately predict R&D expenditures for Switzerland.

Increasing the frequency and consistency of data points could potentially improve the model's performance. In the case of China, the influence of government censorship, commonly referred to as the Great Firewall, has significantly restricted access to Google. This limitation affects the representativeness and reliability of Google trends data, as a substantial portion of internet searches can be routed through local search engines, *e.g.* Baidu (StatCounter, 2024a). For countries like Japan and South Korea, Google competes with other search engines that are more integrated with the respective languages and cultural contexts, *e.g.* Yahoo! in Japan (StatCounter, 2024b) and Naver in South Korea (StatCounter, 2024c). This competition affects the volume and characteristics of data available from Google trends in these countries and how comprehensive the view of search behaviors are. Lastly, a common factor affecting all models' performance for all countries is the varying rate of Google's adoption over time, which directly influences the values obtained from Google trends. Addressing these limitations will further enhance the model's accuracy, leading to more reliable and robust outcomes across different countries.

A separate issue concerns predictive uncertainty. The model-selection ensembles of fifteen independently initialized neural networks allow us to assess the sensitivity of the predictions to random initialization. However, the dispersion across ensemble members does not capture all sources of predictive uncertainty, and intervals based only on this dispersion achieve coverage below their nominal level. Appendix I.F reports the country examples, empirical coverage, and a retrospective recalibration exercise. A separate calibration period or rolling procedure would be required to assess prediction interval coverage for future predictions.

## 5.2 Econometric benchmarks and forecast-comparison tests

Figure 2 benchmarks the neural network against the centered minimum-norm OLS implementation using the same input space. To position the model relative to the econometric nowcasting literature, we further compare it to a set of standard time-series benchmarks evaluated on the identical temporal test set: a no-change random walk (RW), a first-order autoregression (AR(1)), the historical country mean, and two mixed-frequency regressions that map the within-year monthly search index into the annual target, an unrestricted MIDAS (U-MIDAS) and a re-

stricted MIDAS with Beta lag-polynomial weights (Ghysels et al., 2004; Foroni et al., 2015). All Google Trends normalization constants are estimated using the chronological training period only and then held fixed for validation and test observations. Crucially, GERD is released with a publication lag of two to three years (Section 1), so the most recent annual value $y_{t-1}$ is *not* available when nowcasting year $t$ in real time. We therefore report the autoregressive benchmarks at publication-lag-consistent horizons: RW($L$) and AR(1)($L$) use the most recent annual value actually available under a lag of $L$ years, with $L = 3$ being the realistic feasible benchmark (and $L = 1, 2$ shown for reference). We assess statistical significance using the Diebold-Mariano test (Diebold and Mariano, 1995) with the small-sample correction of Harvey et al. (1997), taking the feasible RW($L$=3) as the reference forecast. Table 3 reports the results.

The Diebold-Mariano test evaluates whether the average difference in forecast loss between two models is statistically distinguishable from zero. A negative statistic indicates lower squared-error loss than the reference forecast, while a positive statistic indicates higher loss.

Table 3: Temporal-split out-of-sample accuracy (annual test points, $n = 23$). The neural network rows use models estimated on the combined training and validation periods. $R^2_{\text{TSS}} = 1 - \text{SSE}/\text{TSS}$ is the conventional test-set coefficient of determination. DM tests compare each model with the feasible RW($L$=3) benchmark.

| Model | MAPE (%) | RMSE | $R^2_{\text{TSS}}$ | DM vs RW(3) | $p$ |
|---|---|---|---|---|---|
| *LagRD* | 7.80 | 29.00 | 0.98 | -2.77 | 0.01 |
| *Macros* | 8.54 | 43.85 | 0.96 | -1.15 | 0.26 |
| *AGT* | 7.25 | 36.96 | 0.97 | -2.13 | 0.05 |
| *MGT* | 9.65 | 44.67 | 0.96 | -1.29 | 0.21 |
| *AGTwRD* | 6.82 | 32.77 | 0.98 | -2.34 | 0.03 |
| *MGTwRD* | 8.53 | 38.85 | 0.97 | -1.95 | 0.06 |
| *AllVar* | 8.15 | 34.37 | 0.97 | -2.30 | 0.03 |
| *RW (L=1)* | 4.80 | 19.46 | 0.99 | -2.68 | 0.01 |
| *RW (L=2)* | 8.26 | 38.06 | 0.97 | -2.64 | 0.01 |
| *RW (L=3, feasible)* | 12.04 | 54.94 | 0.93 | | |
| *AR(1) (L=3)* | 5.18 | 19.22 | 0.99 | -2.74 | 0.01 |
| *MIDAS (GT)* | 32.94 | 157.65 | 0.46 | 2.18 | 0.04 |
| *U-MIDAS (GT)* | 34.20 | 163.35 | 0.42 | 2.24 | 0.04 |
| *sg-LASSO-MIDAS (GT, 47 common topics)* | 42.90 | 392.53 | -2.33 | 1.13 | 0.27 |
| *Hist. mean* | 27.29 | 113.68 | 0.72 | 2.65 | 0.01 |

Two points are worth emphasizing. First, R&D expenditure is highly persistent and follows a smooth trend, making simple models based only on its past values difficult to outperform

when predicting annual levels. A random walk using the previous year's value ($L$=1) achieves a 4.8% MAPE, although that value would not generally be available at the forecast date. Even the AR(1) model using GERD delayed by three years ($L$=3), consistent with the publication delay, projects the log linear trend accurately and achieves a 5.2% MAPE. The neural network models do not outperform this autoregressive trend benchmark on annual levels. The contribution of the framework therefore lies in timeliness and temporal resolution rather than in improving annual level forecasts. Several configurations do, however, improve on the feasible no-change benchmark RW($L$=3), which attains 12.0% MAPE. In particular, *AGT*, *AGTwRD*, and *AllVar* attain 7.25%, 6.82%, and 8.15% MAPE, respectively, and have significantly lower squared-error loss than RW($L$=3) at the 5% level. This reflects the structure of the prediction problem rather than a weakness of the search data: when a series follows a strong, smooth trend, timely indicators may provide little additional information beyond past values for predicting its level.

Second, comparisons among models using Google Trends information available during the year being predicted are informative about *where* the neural network's advantage comes from. After applying training-period normalization consistently, the composite-indicator MIDAS and U-MIDAS regressions attain 32.9% and 34.2% MAPE, respectively, with out-of-sample $R^2$ of 0.46 and 0.42. Their squared-error differences relative to *AGT* are statistically significant at the 5% level. The high-dimensional sg-LASSO-MIDAS benchmark (Babii et al., 2022) uses the 47 topics observed across all countries, Legendre polynomial weights to summarize their monthly patterns, and sparse-group regularization to select relevant predictors. It is unstable under the same normalization based only on the training data. It attains 42.9% MAPE, RMSE 392.5, and $R^2 = -2.33$, compared with 7.25%, 37.0, and 0.97 for the *AGT* neural network. Part of this poor performance is due to large extrapolation errors for China. However, the neural network still has a substantially lower MAPE when China is excluded as a diagnostic check. The *AGT* model also has significantly lower squared-error loss than sg-LASSO-MIDAS in the test sample ($p = 0.01$). Thus, for the search-only specification that supplies Step B sensitivities, the neural network is the best-performing search-based benchmark considered across all three reported metrics. This result supports the use of a flexible high-dimensional prediction function, but again does not prove that nonlinearities are required. The sparse-group LASSO assigns nonzero

coefficients to 40 of the 47 common Google Trends topics, indicating limited selection at the topic level. Despite the poor predictive performance of this specification, the Ridge robustness checks show that a regularized linear model can still capture part of the predictive signal. The advantage of the search-only neural network over composite-indicator MIDAS also appears under the alternative country-year grouped split reported in Appendix I.K.

**When does the neural network dominate?** Taken together, Figure 2 and Table 3 show the regime in which the neural network is useful, but also the limits of what we can infer from the present sample. Its advantage over the reported centered minimum-norm OLS is concentrated in the *high-dimensional, densely informative* input spaces: once the 171-228 Google trends features enter the model, the neural network attains a much lower out-of-sample error than this OLS implementation at the same input space, because the unregularized linear fit overfits the many correlated search regressors and extrapolates poorly to future test years. More importantly for the framework, the *AGT* neural network improves on the corrected high-dimensional sg-LASSO-MIDAS benchmark on MAPE, RMSE, and $R^2$. However, this numerical advantage and the poor temporal extrapolation of sg-LASSO-MIDAS do not establish that the underlying relationship must be nonlinear. The competitive Ridge results show that a stabilized linear approximation can capture much of the usable predictive structure in related specifications. We therefore advocate the neural network as a flexible primary specification for exploiting high-frequency, high-dimensional search data, while treating regularized linear models as essential benchmarks. In other applications, or in a larger R&D sample with richer variation, nonlinearities and interactions may become empirically more important; the advantage of the framework is that Step B can use sensitivities from whichever Step A model is most credible.

Two additional Step A diagnostics are reported in the supporting information. Appendix I.C.1 examines performance in smooth and turning-point years, while Appendix I.C.2 presents the global SHAP interpretability exercise.

# 6 Interpolating R&D expenditures at a higher frequency

Google Trends data are available at a higher frequency than annual GERD. We use this granularity to construct monthly estimates through temporal disaggregation, namely the use of high-frequency indicators to distribute observed low-frequency totals across subperiods (Mosley et al., 2022). This exercise is distinct from an operational monthly nowcast. Here, the annual GERD total and the complete set of monthly search observations within that year are observed, and Step B estimates the unobserved within-year allocation. If the annual total is not yet available, it may instead be replaced by the Step A nowcast, as discussed below.

## 6.1 The temporal-disaggregation problem

Suppressing the country index temporarily, let $\mathbf{y}^{(m)} \in \mathbb{R}^{12n}$ collect the unobserved monthly GERD values for $n$ years and let $\mathbf{y} \in \mathbb{R}^n$ collect the corresponding observed annual totals. Throughout this subsection, indexed lowercase quantities denote individual annual or monthly observations, while bold lowercase quantities denote the corresponding series stacked over time. The uppercase vector $\mathbf{Y}_{t-\tau,i}$ used in Step A continues to denote the finite vector of lagged GERD inputs. Define the aggregation matrix

$$\mathbf{A} = \mathbf{I}_n \otimes \mathbf{1}_{12}^{\top}, \qquad \mathbf{y} = \mathbf{A}\mathbf{y}^{(m)}. \tag{2}$$

Each row of $\mathbf{A}$ sums the twelve monthly values associated with one year, corresponding to $y_{t,i} = \sum_{j=0}^{11} y_{t,j,i}^{(m)}$. For $n$ years, equation (2) provides only $n$ annual aggregation restrictions for $12n$ unknown monthly values. The monthly path is therefore not identified from annual GERD alone. Temporal-disaggregation methods must supply additional information through high-frequency indicators, assumptions about residual dynamics or smoothness, or a combination of these elements. We consider three approaches: the regression-based method of Chow and Lin (1971), its sparse extension for high-dimensional indicator sets proposed by Mosley et al. (2022), and our model-implied sensitivity allocation.

## 6.2 Regression-based temporal disaggregation

The regression-based methods begin from the latent monthly model

$$\mathbf{y}^{(m)} = \mathbf{X}^{(m)}\beta + \mathbf{u}^{(m)}, \qquad \mathbb{E}[\mathbf{u}^{(m)}] = \mathbf{0}, \qquad \mathrm{Var}(\mathbf{u}^{(m)}) = \mathbf{V}_m. \tag{3}$$

Here, $\mathbf{X}^{(m)} \in \mathbb{R}^{12n\times d}$ is the monthly design matrix containing $d$ high-frequency indicators, $\beta \in \mathbb{R}^d$ is the corresponding vector of regression coefficients, and $\mathbf{u}^{(m)} \in \mathbb{R}^{12n}$ is the vector of monthly disturbances. We reserve $\beta$ for the coefficients of the regression-based temporal-disaggregation models, while $\theta^{(m)}$ continues to denote the parameters of the Step A prediction function. In the Chow-Lin specification, the monthly disturbances follow an AR(1) process with parameter $|\rho| < 1$. Aggregating equation (3) gives

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}, \qquad \mathbf{X} = \mathbf{A}\mathbf{X}^{(m)}, \qquad \mathbf{u} = \mathbf{A}\mathbf{u}^{(m)}, \qquad \mathbf{V}_a = \mathbf{A}\mathbf{V}_m\mathbf{A}^\top. \tag{4}$$

For a given value of $\rho$, the classical estimator is obtained by GLS. The sparse extension of Mosley et al. (2022) adds a penalty to permit estimation when the number of indicators is large:

$$\hat{\beta}_\rho = \arg\min_{\beta\in\mathbb{R}^d} \left\{ \left\| \mathbf{V}_a^{-1/2}(\mathbf{y} - \mathbf{X}\beta) \right\|_2^2 + P_\lambda(\beta) \right\}, \tag{5}$$

where $P_\lambda(\beta) = 0$ gives classical Chow-Lin and $P_\lambda(\beta) = \lambda\|\beta\|_1$ gives the sparse temporal-disaggregation estimator considered here. After estimating the regression signal, the Chow-Lin best linear unbiased predictor distributes the remaining annual discrepancy according to the estimated monthly covariance structure:

$$\hat{\mathbf{y}}_{\mathrm{CL}}^{(m)} = \mathbf{X}^{(m)}\hat{\beta} + \mathbf{V}_m\mathbf{A}^\top(\mathbf{A}\mathbf{V}_m\mathbf{A}^\top)^{-1}\left(\mathbf{y} - \mathbf{A}\mathbf{X}^{(m)}\hat{\beta}\right). \tag{6}$$

Equation (6) separates the fitted high-frequency indicator component from the covariance-based reconciliation of the annual regression residual. It also satisfies $\mathbf{A}\hat{\mathbf{y}}_{\mathrm{CL}}^{(m)} = \mathbf{y}$. We implement the *chow-lin-maxlog* method using the *tempdisagg* R package (Sax and Steiner, 2013). For the sparse estimator of Mosley et al. (2022), we use the *DisaggregateTS* R package (Mosley and

Nobari, 2022).

## 6.3 Model-implied sensitivity allocation

The proposed estimator uses the same accounting constraint but constructs the high-frequency indicator signal differently. We use *model-implied sensitivity allocation* as the umbrella framework label for weighting the realized monthly timing profiles of the full set of Google Trends topics. The main specification measures topic relevance through small Gaussian perturbations to the standardized model inputs. The neural network is the primary implementation because it can accommodate high-dimensional inputs, interactions, and nonlinearities when supported by the data, while the allocation rule itself is model-agnostic.

The current setup uses the *AGT* model from Step A and conducts a separate sensitivity analysis for Step B. The main text reports Gaussian perturbation relevance with a perturbation standard deviation of 0.10 on the standardized input scale. The resulting monthly path is effectively unchanged when the standard deviation varies from 0.05 to 0.25. Appendix I.I.1 reports the earlier deterministic sensitivity definitions as additional checks. The correlations below describe how the estimates compare across specifications and methods, but they cannot validate the unobserved monthly GERD path.

For topic $s_{k_s}$, lag $\tau$, year $t$, monthly information row $j$, and country $i$, let $\hat{y}_{t,i|j} = f^{(m)}(\mathbf{x}_{t,j,i}; \hat{\theta}^{(m)})$ denote the fitted Step A prediction and let $\hat{y}^{\delta}_{t,i|j} = f^{(m)}(\mathbf{x}^{\delta}_{t,j,i}; \hat{\theta}^{(m)})$ denote the prediction after adding $\delta$ to one standardized search-topic input. The notation $\hat{y}_{t,i|j}$ emphasizes that the Step A target remains annual GERD for country $i$ in year $t$, while $j$ indexes the monthly information row used to form the prediction. It is distinct from $\hat{y}^{(m)}_{t,j,i}$, which denotes estimated monthly GERD only after temporal disaggregation in Step B. We define Gaussian perturbation relevance as

$$\hat{\eta}_{k_s,\tau,t,j,i}(\hat{\theta}^{(m)}) = \frac{1}{\sigma \mathbb{E}|Z|} \mathbb{E}_{\delta \sim \mathcal{N}(0,\sigma^2)} \left[ \left| \frac{\hat{y}^{\delta}_{t,i|j} - \hat{y}_{t,i|j}}{\hat{y}_{t,i|j}} \right| \right], \qquad Z \sim \mathcal{N}(0,1). \tag{7}$$

The perturbation changes one annual search-topic input while holding the remaining inputs and the fitted parameter vector $\hat{\theta}^{(m)}$ fixed. We use $\sigma = 0.10$ and 50 antithetic pairs, so each positive draw is matched by its negative. Dividing by $\sigma \mathbb{E}|Z|$ expresses the expected absolute relative response per standardized-input unit. This relevance measure summarizes the

combined sensitivity generated by all neural network weights, biases, activation functions, and interactions, rather than representing one topic-specific network coefficient. We average over the training observations and neural network ensemble members. Because the current model contains three annual representations of each topic, the implementation first estimates a separate relevance measure for each lag and then averages those lag-specific values to obtain a stable country-topic weight:

$$\bar{\eta}_{k_s,i} = \frac{1}{|\mathscr{L}_{k_s}|} \sum_{\tau \in \mathscr{L}_{k_s}} \mathbb{E}_{S_{\text{train}}}[\hat{\eta}_{k_s,\tau,t,j,i}(\hat{\theta}^{(m)})], \qquad \mathscr{L}_{k_s} = \{1, 2, 3\}. \tag{8}$$

Thus, the nonlinear Step A function is used to estimate predictive relevance, while the Step B topic weights are deliberately stabilized across perturbations, observations, ensemble members, and lags. These quantities are reduced-form predictive sensitivities under the model and information set. They are not structural or causal elasticities.

Let the monthly share of topic $s_{k_s}$ within year $t$ be

$$p^{(m)}_{s_{k_s},t,j,i} = \frac{s^{(m)}_{k_s,t,j,i}}{\sum_{h=0}^{11} s^{(m)}_{k_s,t,h,i}}, \qquad \sum_{j=0}^{11} p^{(m)}_{s_{k_s},t,j,i} = 1. \tag{9}$$

We combine predictive relevance and within-year timing through the monthly score

$$g_{t,j,i} = \sum_{k_s} \bar{\eta}_{k_s,i} p^{(m)}_{s_{k_s},t,j,i}, \tag{10}$$

and normalize this score within each country-year:

$$q_{t,j,i} = \frac{g_{t,j,i}}{\sum_{h=0}^{11} g_{t,h,i}}, \qquad \hat{y}^{(m)}_{t,j,i} = y_{t,i} q_{t,j,i}. \tag{11}$$

Since every topic profile sums to one, the denominator in equation (11) equals $\sum_{k_s} \bar{\eta}_{k_s,i}$. The allocation is therefore well defined when this annual timing-score denominator is nonzero and the resulting monthly scores are admissible. The resulting estimates are positive in every month in our sample, but positivity is an empirical property rather than a general mathematical guarantee. If negative allocations arose in another application, the estimator would require sign

constraints, a positive transformation of the scores, or constrained benchmarking. Provided that the denominator is nonzero, equation (11) satisfies the annual accounting constraint exactly:

$$\sum_{j=0}^{11} \hat{y}_{t,j,i}^{(m)} = y_{t,i}.$$

When the annual total is not yet observed, $y_{t,i}$ may be replaced by the Step A nowcast $\hat{y}_{t,i}$, although the resulting operational high-frequency nowcast contains uncertainty from both steps and requires separate validation.

The allocation has a transparent interpretation. Annual GERD provides the scale, the monthly search profiles provide the timing, and the model-implied sensitivities determine the relative predictive relevance of the topics. If an influential topic is unusually concentrated in a particular month, that month receives a larger share of the annual total. The framework consequently accommodates fixed and evolving seasonal information in different ways. Month indicators in the prediction model control average calendar effects as fixed categorical inputs, while Step B uses realized country-year-topic search shares that can change across years. We do not estimate a fully parametric monthly GERD seasonal process because monthly GERD is unobserved and the annual sample is small. Additional month-specific or time-varying seasonal parameters would therefore be weakly identified and could fit noise rather than genuine within-year variation.

## 6.4 Relationship with regression-based estimators

The proposed method is best viewed as model-implied construction of a high-frequency indicator followed by exact proportional benchmarking to annual GERD. Its connection with regression-based temporal disaggregation can be shown using the original-scale elasticity variant and specializing the Step A prediction function to a linear map. If

$$f(\bar{\mathbf{s}}_{t,i}; \beta) = \sum_{k_s} \beta_{k_s} \bar{s}_{k_s,t,i},$$

then the elasticity of topic $k_s$ is $\eta_{k_s,t,i} = \beta_{k_s}\bar{s}_{k_s,t,i}/f(\bar{\mathbf{s}}_{t,i};\beta)$. Using $\sum_h s^{(m)}_{k_s,t,h,i} = 12\bar{s}_{k_s,t,i}$ gives

$$\eta_{k_s,t,i}p^{(m)}_{s_{k_s},t,j,i} = \frac{\beta_{k_s}s^{(m)}_{k_s,t,j,i}}{12f(\bar{\mathbf{s}}_{t,i};\beta)},$$
$$\hat{y}^{(m)}_{t,j,i} = y_{t,i}\frac{\mathbf{x}^{(m)\top}_{t,j,i}\beta}{\sum_{h=0}^{11}\mathbf{x}^{(m)\top}_{t,h,i}\beta}.$$

Thus, in the linear case the elasticity-weighted timing score is proportional to the fitted linear indicator $\mathbf{x}^{(m)\top}_{t,j,i}\beta$. The proposed estimator then reduces to proportional benchmarking based on that fitted indicator. This establishes a direct connection to the indicator component of Chow-Lin and sparse temporal disaggregation, but not an equivalence with the complete Chow-Lin predictor in equation (6). Chow-Lin additionally allocates the annual regression residual using an estimated covariance structure, whereas our method normalizes the model-implied timing signal directly.

This comparison clarifies both the generalization and its limits. The Step A predictor may estimate topic relevance through a nonlinear, high-dimensional function, but the principal Step B implementation uses stabilized average Gaussian perturbation relevance to form the monthly timing score. Regularized linear robustness checks therefore remain informative. Under the same Gaussian perturbation procedure, the Ridge and neural network allocations are highly correlated ($r = 0.997$ in levels and $r = 0.989$ in monthly growth). This indicates that much of the usable signal in the present sample can be represented by a stabilized linear approximation. At the same time, differences across models show that the exact within-year path depends on the predictive-sensitivity estimator. The framework retains the capacity to accommodate interactions and nonlinearities in applications with richer data.

The estimator rests on three requirements. First, the monthly estimates must sum to the corresponding annual total. Second, predictive sensitivities estimated from annual data must provide informative relative weights for allocating that total across months. This assumes that relationships learned from annual variation contain information about within-year allocation; it does not treat annual sensitivities as structural monthly parameters (Woloszko, 2020). Third, the observed monthly search shares must contain information about the timing of R&D-related

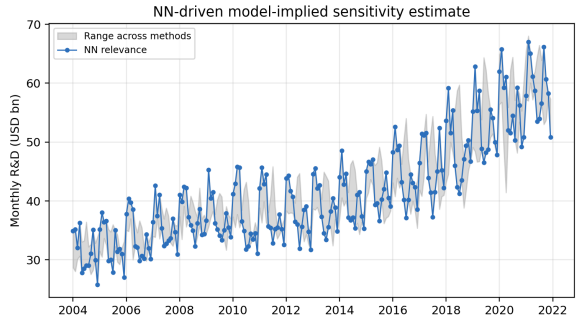
activity. Because annual observations cannot reveal the unobserved monthly path, the method produces an allocation consistent with the selected indicators rather than an estimate proven to recover the true monthly GERD series. We assess its plausibility and robustness by comparing it with estimates from other temporal disaggregation methods, using alternative Step A models, aggregating the estimates to quarters, and comparing them with observed employment in R&D services. The monthly estimates should primarily be used for quarterly or seasonal monitoring, and isolated monthly movements should not be overinterpreted.[8]

When the prediction model uses standardized predictors, relevance can be measured on either the original or standardized scale and with deterministic or stochastic perturbations. These quantities are not algebraically equivalent, but each can be used in equations (10) and (11). They preserve the annual total when the denominator of the monthly timing score is nonzero. The main results use Gaussian perturbations on the standardized input scale. Appendix I.I.1 reports the earlier original-scale elasticity and deterministic standardized-input results. Correlations with estimates from other temporal disaggregation methods are used as descriptive comparisons, not as evidence that one definition recovers the unobserved monthly GERD path more accurately.

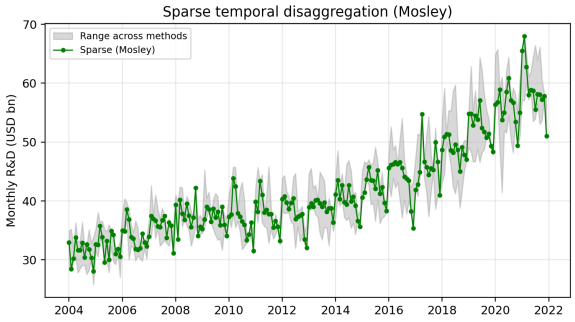
For the Chow-Lin comparison, we use six SHAP-informed indicators.

Figure 3 depicts different estimated monthly R&D expenditures over time for the United States. The neural network estimate uses Gaussian perturbation relevance from the Step A *AGT* model. It is strongly correlated with the sparse temporal-disaggregation estimate (Mosley et al., 2022), with $r(214) = 0.91$, $p < 0.001$ in levels, $r(213) = 0.68$, $p < 0.001$ in monthly growth, and $r(69) = 0.74$, $p < 0.001$ after quarterly aggregation. The resulting monthly path is effectively unchanged when the perturbation standard deviation varies from 0.05 to 0.25. We also apply the same Gaussian relevance procedure to centered minimum-norm OLS and validation-tuned Ridge models estimated using the same sample. Their allocations are highly correlated with the neural network allocation: $r = 0.997$ and $r = 0.997$ in levels, and $r = 0.984$ and $r = 0.989$ in monthly growth, for OLS and Ridge respectively. These results support a

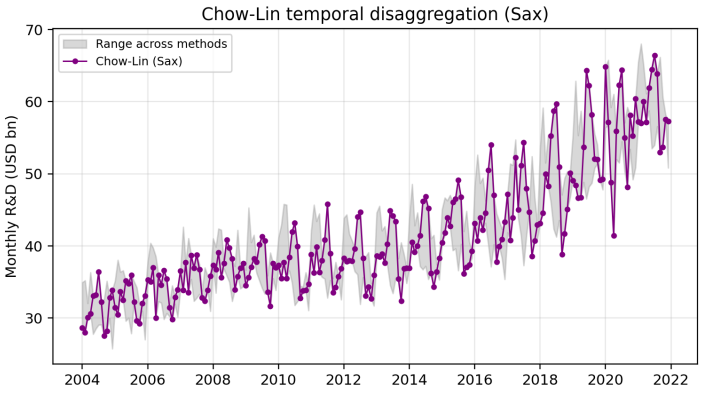
[8]Appendix I.J presents an alternative implementation based on perturbing the model inputs. Because monthly GERD is unobserved, it is reported as a supplementary construction rather than as a method that can be ranked against the sensitivity-based procedure using monthly ground truth.




(a) Model-implied sensitivity allocation




(b) Sparse temporal disaggregation




(c) Chow–Lin temporal disaggregation

Figure 3: Comparison of Estimated Monthly R&D Expenditures for the United States. Panel (a) uses Gaussian perturbation relevance from the Step A *AGT* model, panel (b) uses sparse temporal disaggregation Mosley et al. (2022), and panel (c) uses the method of Chow and Lin (1971).

model-agnostic interpretation of Step B: the current monthly signal is not an artifact of one neural network specification, and in this small sample much of the usable predictive structure can be approximated by a linear model. The classical temporal-disaggregation method of Chow and Lin (1971) uses the six SHAP-informed indicators. The correlation between the Chow–Lin estimate shown in Figure 3c and the neural network model-implied sensitivity estimate is $r(214) = 0.69$, $p < 0.001$ in levels and $r(213) = -0.26$, $p < 0.001$ in monthly growth. After quarterly aggregation, the correlation is $r(69) = 0.21$, $p = 0.083$.[9]

We note that, because all three methods disaggregate the *same* observed annual totals, the twelve monthly figures are constrained to sum to the corresponding yearly value; the level series therefore share a common low-frequency envelope (the annual totals and their

[9]The correlation between the temporal disaggregation shown in Figure 3c and Figure 3b is $r(214) = 0.87$, $p < 0.001$ in level, and $r(213) = 0.17$, $p = 0.011$ on the growth rates.

trend), so the level correlations are in part mechanical. The correlation in *growth rates* is consequently more informative because it is more sensitive to differences in within-year allocation across methods. Through this lens, the two full-information methods (the neural network-driven estimates and the sparse method of Mosley et al. (2022)) are strongly correlated in the high-frequency dynamics ($r = 0.68$ in monthly growth, rising to $r = 0.74$ after quarterly aggregation). The classical Chow and Lin (1971) method, restricted to six indicators, captures a different within-year pattern: the monthly growth correlation is negative, while the quarterly correlation is positive but not statistically significant.

An exploratory external comparison with employment in scientific R&D services is reported in Appendix I.H. This auxiliary exercise is separate from the Gaussian relevance comparison above and is not used to validate or rank the temporal-disaggregation methods.

# 7 Conclusion and future work

This study develops a two-step, prediction-oriented framework for improving the timeliness and frequency of GERD measurement. Step A nowcasts annual GERD using mixed-frequency information, with particular emphasis on Internet search intensity. Step B uses model-implied predictive sensitivities and within-year search profiles to allocate annual GERD across months. The framework therefore addresses two related limitations of official R&D statistics: delayed publication and low reporting frequency. It is designed for real-time monitoring and measurement, not for causal inference or policy counterfactual analysis.

When the models are fitted on earlier country-years and evaluated on later ones, Google Trends configurations are competitive with configurations using traditional macroeconomic predictors. In the search-only input space used for temporal disaggregation, the *AGT* neural network records a lower MAPE than the macro-only model and the linear and mixed-frequency benchmarks considered using comparable search information. The country-year grouped exercise supports the same qualitative interpretation, although the temporal split remains the primary design for evaluating predictions on later years. At the same time, annual GERD is highly persistent, allowing simple autoregressive benchmarks to achieve strong predictive

accuracy on annual levels. The contribution of Step A should therefore be understood primarily in terms of extracting timely predictive information from alternative data and providing the predictive sensitivities required for Step B, rather than consistently outperforming persistence-based annual forecasts.

The neural network provides a flexible way to accommodate high-dimensional inputs, interactions, and potential nonlinearities. However, the present results do not establish that nonlinear relationships are necessary. Ridge captures part of the predictive signal, and Ridge-based Gaussian relevance produces a monthly allocation that is closely aligned with the one derived from the neural network. The neural network has significantly lower squared-error loss than the regularized MIDAS benchmark in the small annual test sample, although the comparison remains based on only 23 country-year observations.

The neural network is therefore our primary implementation rather than a defining feature of the framework. Step B can use sensitivities from alternative prediction models in Step A. Each model must first perform the prediction task at an acceptable level relative to the benchmarks. The resulting allocation weights must then be assessed for stability and economic plausibility. Appendix I.K shows that the qualitative conclusions remain similar under a grouped split, which keeps all observations from each combination of country and year together but does not preserve temporal order across the training, validation, and test sets. It also documents the information leakage caused by randomly splitting individual monthly observations.

The framework is not designed to address the policy evaluation problem raised by the Lucas critique (Lucas, 1976). Econometric models used for policy evaluation may fail after policy regimes change because agents adjust their expectations and decisions, causing structural parameters to change. Our framework is designed for predictive nowcasting and timely measurement rather than for policy counterfactual analysis. We therefore do not interpret its estimated relationships as structural or behavioral parameters that remain unchanged across policy regimes. The model nevertheless requires relationships learned from earlier observations to remain useful for later predictions. Following the distinction made by Mullainathan and Spiess (2017), our objective is to predict outcomes ($\hat{y}$) under a given information set, rather than to identify causal parameters ($\hat{\beta}$). Search terms related to policy are used as timely indicators

of attention, expectations, or awareness of R&D topics, not as measures of policy instruments or causal treatments.

The estimated monthly series are strongly correlated in levels and growth with the sparse temporal disaggregation estimates, while the comparison with Chow–Lin gives a different within-year pattern. The comparison with employment in scientific R&D services is exploratory because that series covers only one sector and does not measure all R&D personnel, labor costs, or non-labor GERD. Uniform allocation also produces sizeable lagged correlations because annual totals change at year boundaries, and none of the largest lagged correlations is statistically significant under the simultaneous year block test. After annual levels and average month effects are removed, the employment data do not show a robust timing advantage for the neural network or sparse allocation over uniform shares. Because true monthly GERD is unobserved, neither correlations across the estimated series nor correlations with employment identify the most accurate disaggregation method. The estimates should therefore be used cautiously, with greater emphasis on quarterly or seasonal patterns than on individual monthly movements. Moreover, Step B currently disaggregates observed annual totals retrospectively. Combining it with Step A could support an operational monthly nowcasting system, but this extension requires further evaluation.

Several extensions follow naturally. Future research could assess the estimates using additional high-frequency indicators, countries, and periods, particularly around economic shocks and policy changes. Forecast uncertainty could be assessed more fully by incorporating residual uncertainty and using a rolling calibration sample, a separate calibration period, or conformal methods. Larger samples would permit a more decisive assessment of nonlinearities, lag structures, and seasonal dynamics. In countries where Google has limited search market penetration, combining Google Trends with data from locally important search platforms could also improve the representativeness of the underlying indicators.

More broadly, the framework could be applied to less persistent macroeconomic variables, regional data, or other innovation-related activities for which conventional statistics are delayed or available only at low frequency. Such extensions require alternative indicators that are sufficiently representative at the intended geographic and temporal resolution. Very fine

disaggregation may not be credible when web search intensity is sparse, unrepresentative, or only weakly connected to the timing of the underlying activity.

The framework provides a basis for combining annual nowcasting with higher-frequency measurement. With further evaluation, such indicators could complement official statistics and innovation scoreboards by offering earlier evidence on changes in R&D activity, especially during periods when timely monitoring matters most.

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